POWER BI - WEEK 12 – Report

INTRODUCTION

The Week 12 activity focuses on developing an interactive Automobile Analytics Dashboard using Power BI and the preprocessed Automobile Dataset from Week 11.

The dashboard helps an automobile company compare vehicles based on price, engine size, horsepower, fuel efficiency, manufacturers, body styles, and drive wheels.

Various visualizations, including scatter plots, column charts, matrix grids, and KPI cards, are used to analyze automobile performance and market segments. Interactive tooltips are also created to display detailed technical specifications of vehicles.

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Figure 1: Loading the Preprocessed Automobile Dataset and Preparing the Dashboard in Power BI

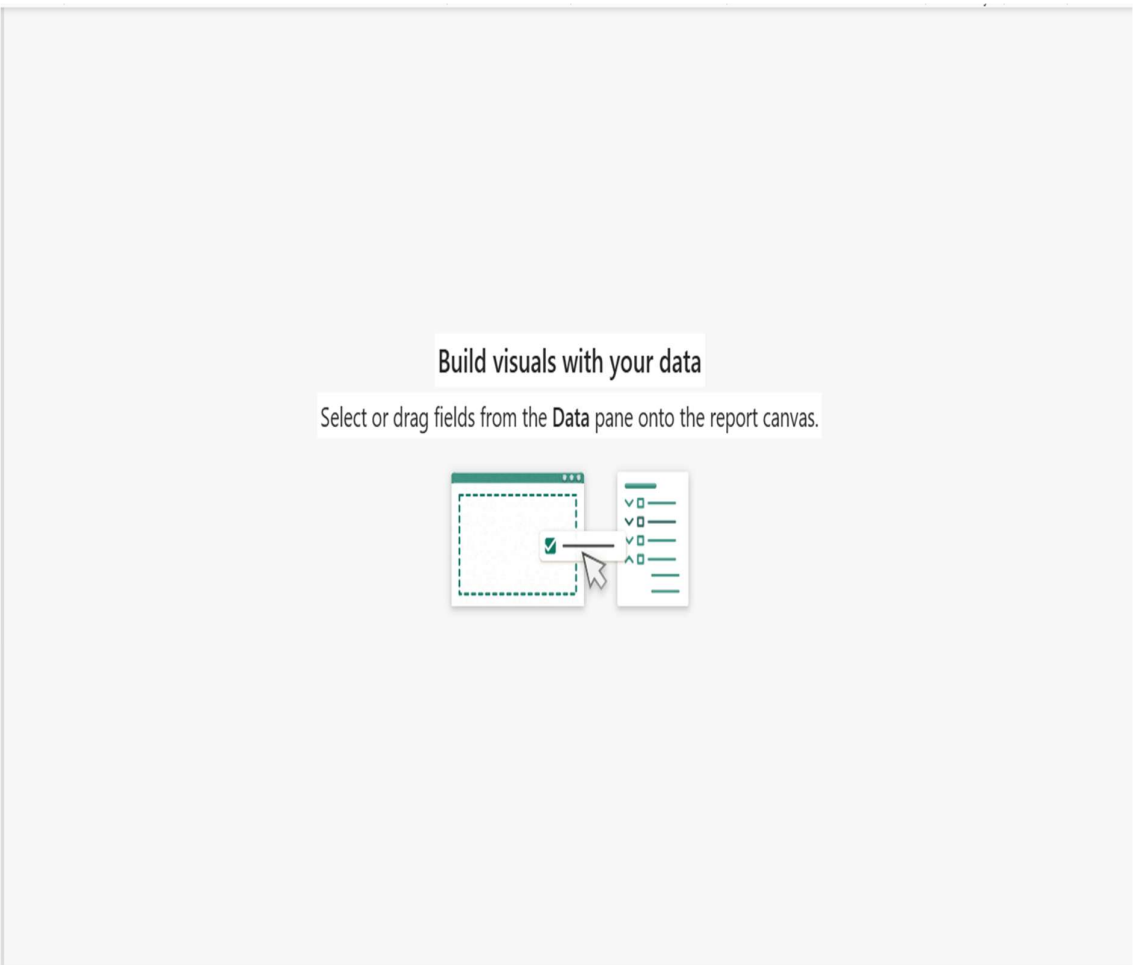


Figure 2: Creating KPI Cards for Average Price, Horsepower, Engine Size, and Highway MPG in Power BI

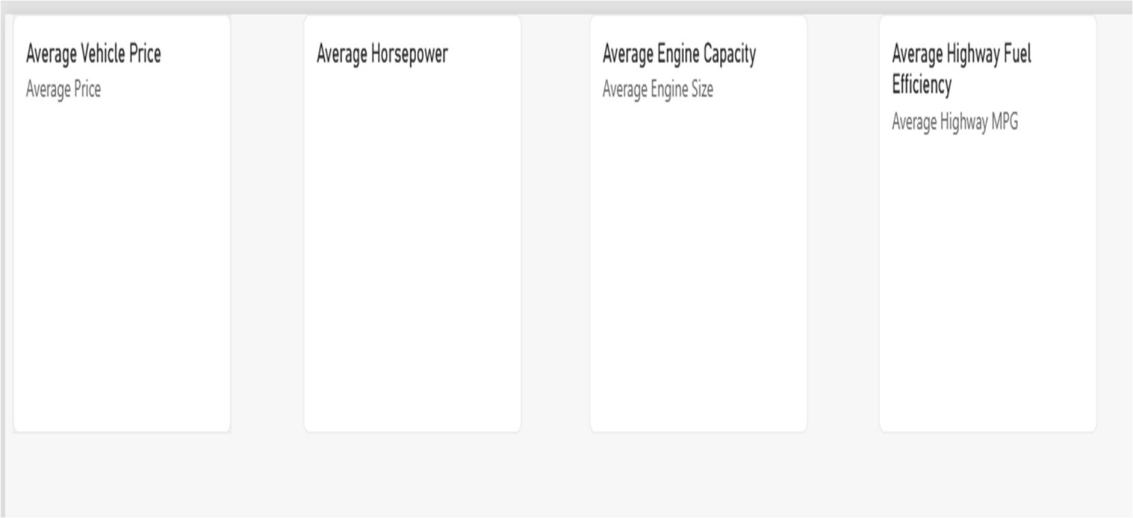


Figure 3: Analyzing Horsepower vs Vehicle Price Using Engine Size Bubble Scaling in Power BI

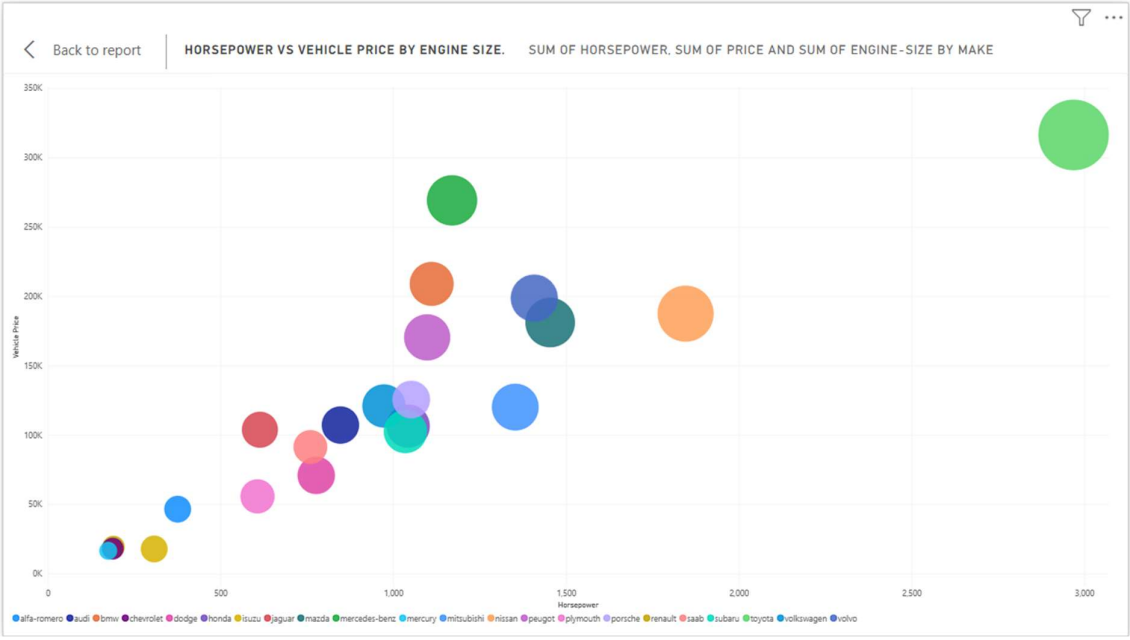


Figure 4: Comparing Average City and Highway Fuel Efficiency Across Manufacturers in Power BI

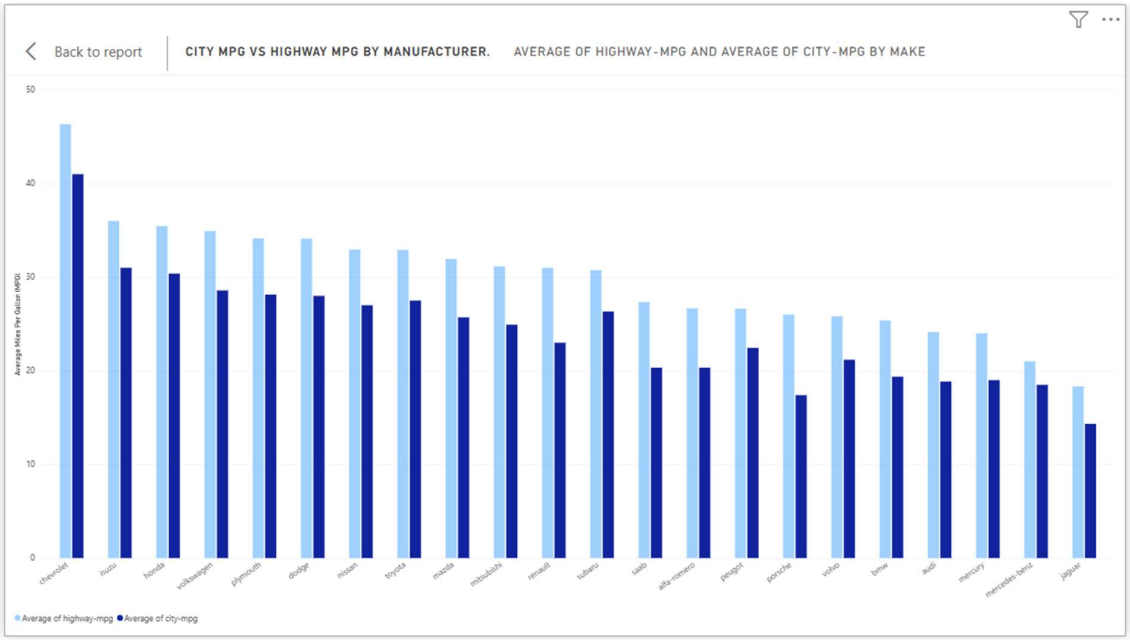


Figure 5: Analyzing the Relationship Between Engine Size and Horsepower Across Manufacturers in Power BI

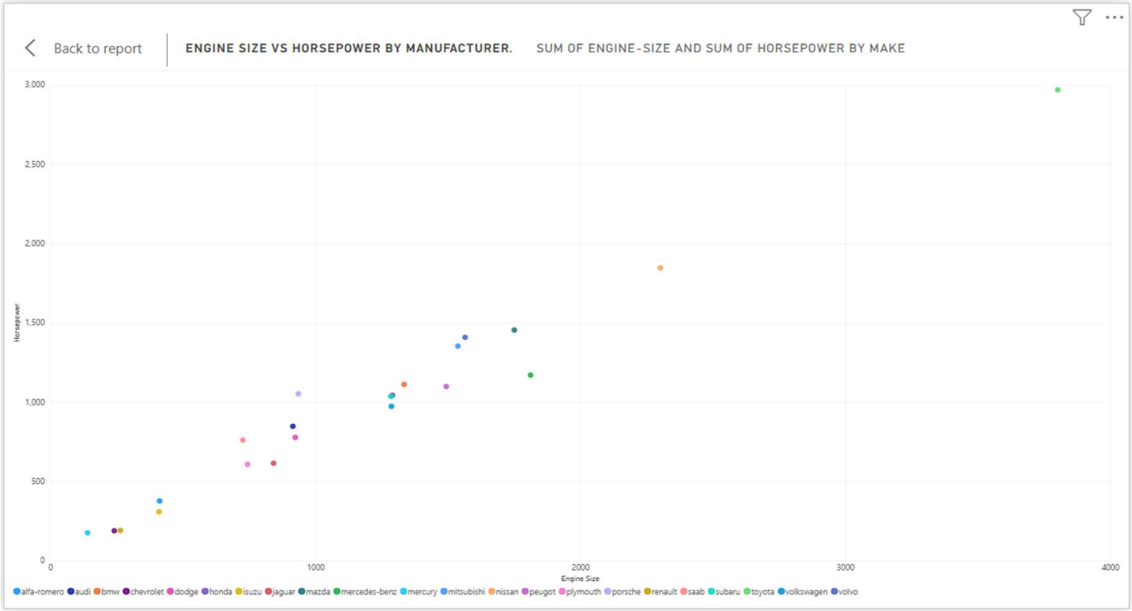
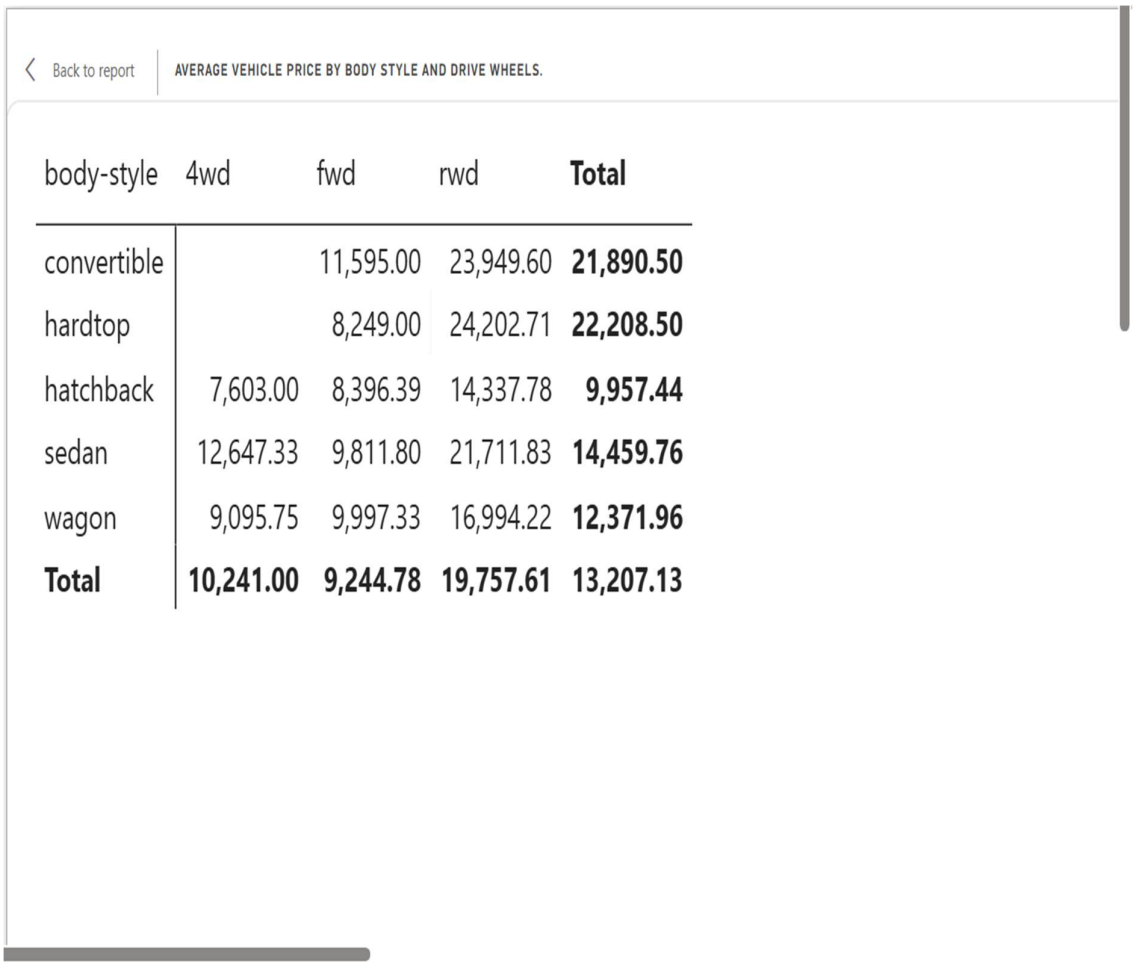


Figure 6: Comparing Average Vehicle Prices Across Body Styles and Drive Wheel Categories Using a Matrix in Power BI



body-style	4wd	fwd	rwd	Total
convertible		11,595.00	23,949.60	**21,890.50**
hardtop		8,249.00	24,202.71	**22,208.50**
hatchback	7,603.00	8,396.39	14,337.78	**9,957.44**
sedan	12,647.33	9,811.80	21,711.83	**14,459.76**
wagon	9,095.75	9,997.33	16,994.22	**12,371.96**
Total	**10,241.00**	**9,244.78**	**19,757.61**	13,207.13

Figure 7: Comparing Automobile Manufacturers Across Budget, Mid-Range, and Luxury Market Segments in Power BI

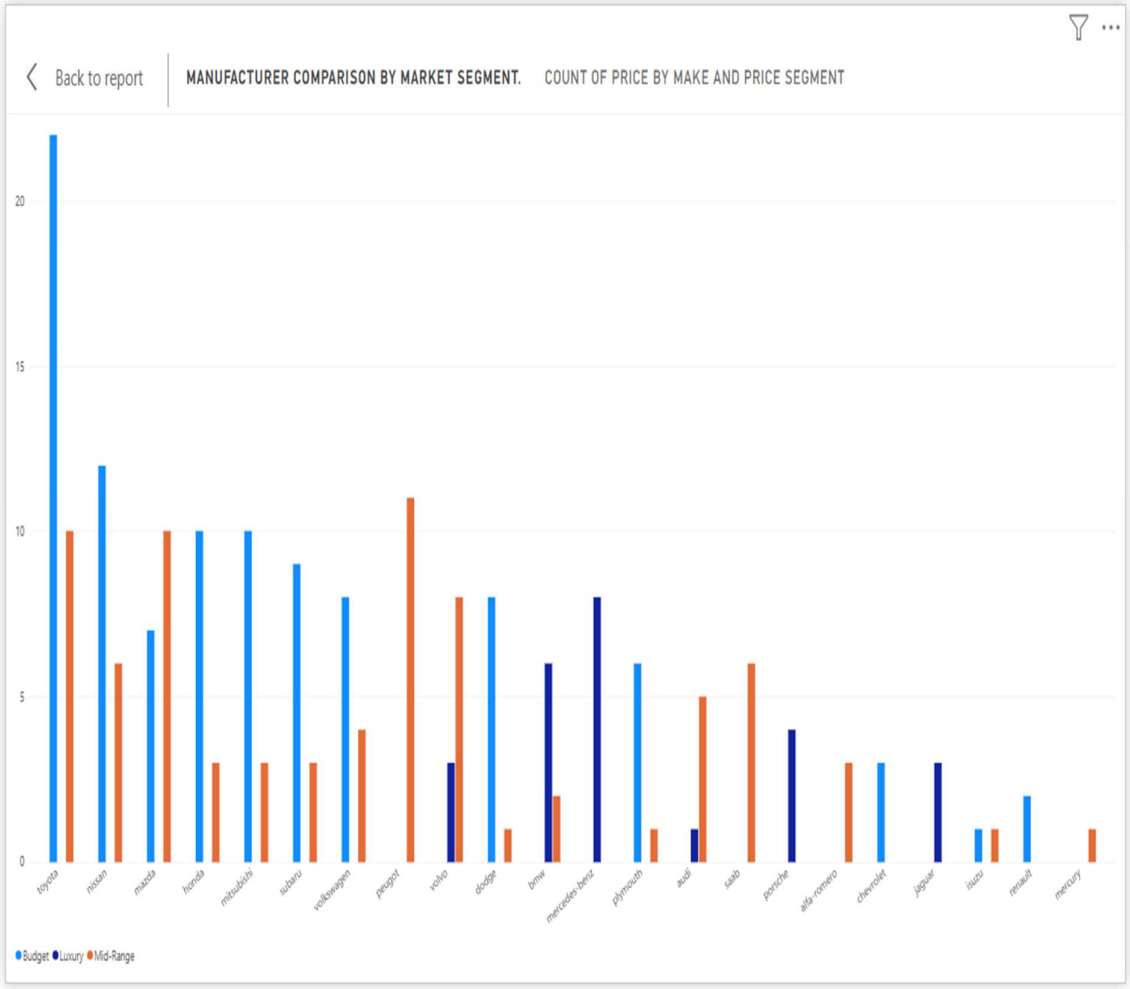
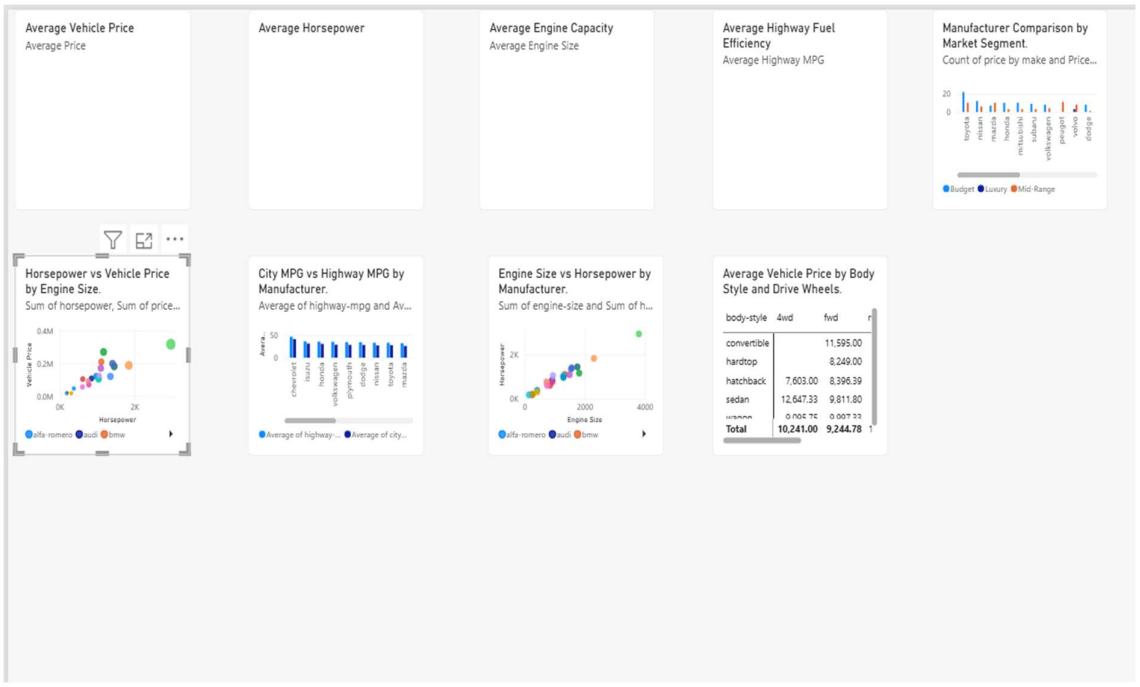


Figure 8: Creating a Custom Vehicle Technical Specifications Tooltip Page in Power BI



Figure 9: Configuring and Testing Interactive Vehicle Technical Tooltips in Power BI



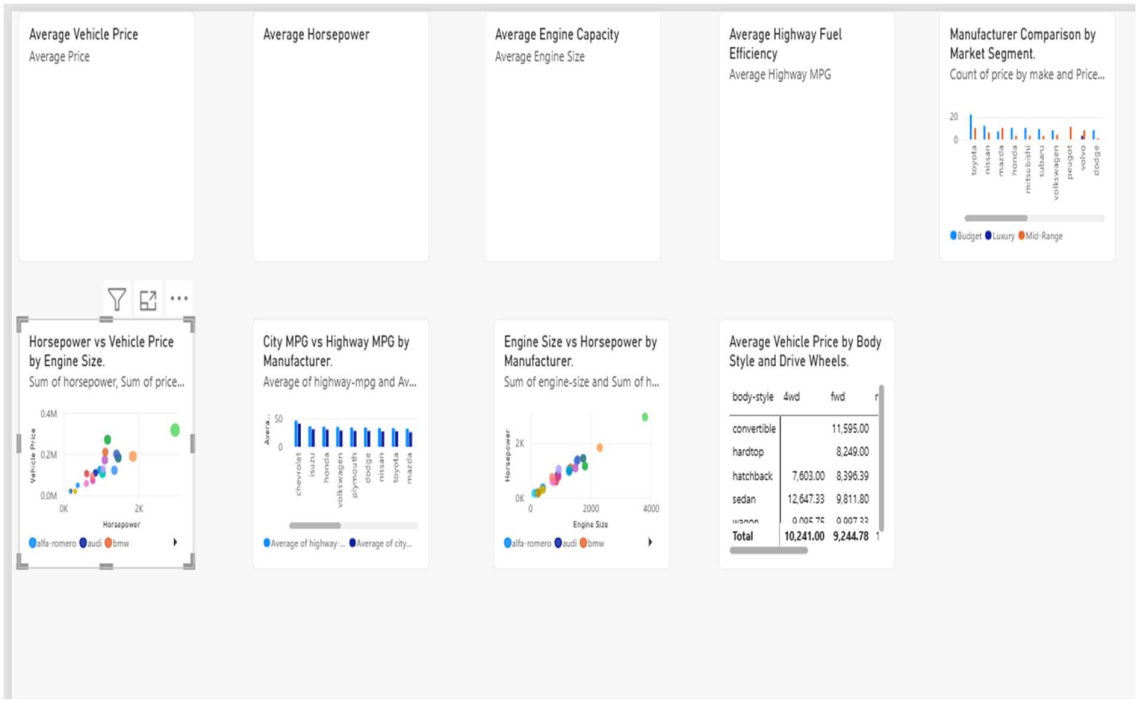
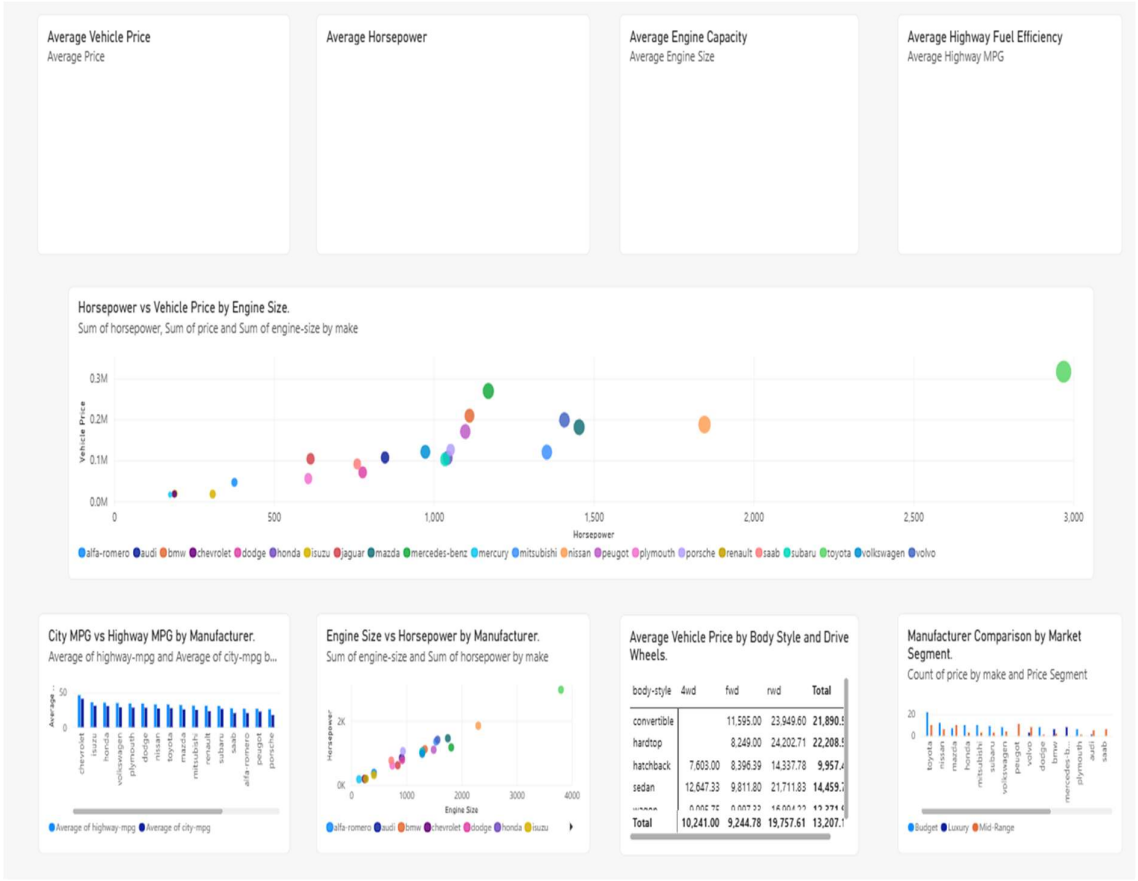
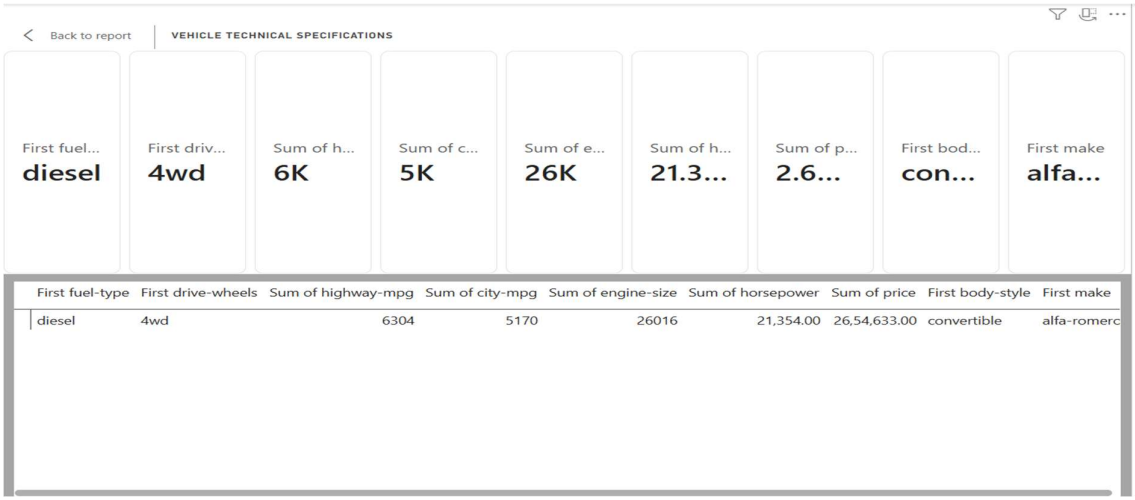


Figure 10: Final Automobile Analytics Dashboard with KPI Cards, Comparative Charts, Matrix Analysis, and Interactive Technical Tooltips in Power BI





Conclusion

The Automobile Analytics Dashboard was successfully created using Power BI. The dashboard presents automobile data through KPI cards, scatter plots, column charts, and a comparison matrix. It helps analyze vehicle prices, horsepower, engine size, fuel efficiency, body styles, drive wheels, and manufacturer market segments.

This project improved my understanding of data visualization, comparative analysis, and dashboard development using Power BI.